



UNIVERSITY OF KELANIYA - SRI LANKA
FACULTY OF SCIENCE

Bachelor of Science Degree Examination, October/November - 2023

Academic Year: 2021/2022 - Semester I

ELECTRONICS AND COMPUTER SCIENCE

BECS 21413 - Analogue Electronics II (Operational Amplifiers)

No. of Questions: Seven (07) No. of Pages: Four (04) Time: 02 hours & 30 minutes

Answer any **five (05)** questions.

(All the symbols and abbreviations have their usual meanings.)

- 1) Write short notes on the following op-amp parameters.
 - (a) The common-mode rejection ratio (CMRR).
 - (b) Output voltage saturation.
 - (c) The input and output impedance.
 - (d) The frequency response.
 - (e) The slew rate.

- 2)
 - (a) Describe the op-amp's negative feedback concept using a suitable diagram.
 - (b) Using suitable circuit diagrams, derive expressions for the output voltage (V_o) of op-amp's basic inverting and non-inverting closed-loop configurations. Using the derived expressions, obtain closed-loop gains for inverting and non-inverting configurations.
 - (c) Using suitable circuit diagrams and the expressions you have obtained in part (b), discuss the construction of inverter and voltage follower circuits.
 - (d) Figure 1 shows a circuit diagram of an AC inverting amplifier with a single power supply.
 - (i) Draw the DC equivalent circuit of the amplifier and show that the output AC signal swings about the DC level of $V_B/2$.
 - (ii) Draw the AC equivalent circuit of the amplifier and show that only the AC signal appears across the load R_L .

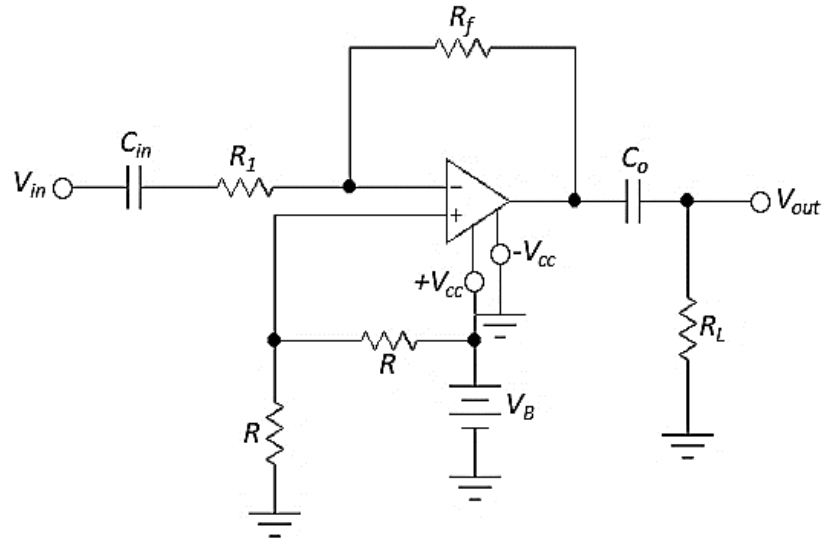


Figure 1

- 3) The circuit diagram of a phase shifter is shown in Figure 2.

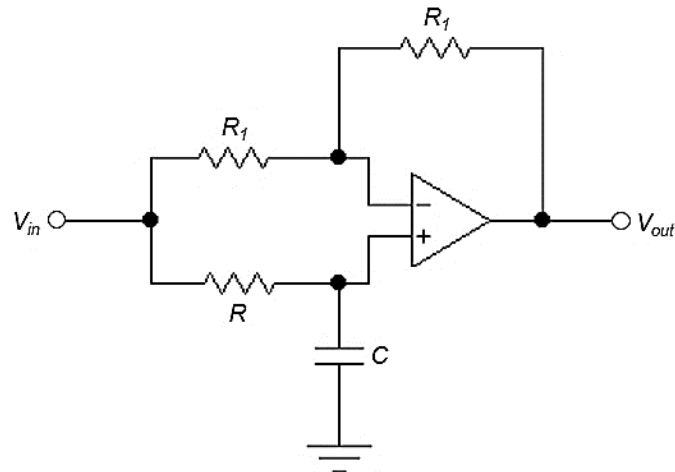


Figure 2

- (a) By applying the superposition theorem, show that the transfer function of the above phase shifter is given by,

$$T(jf) = \frac{1 - j\left(\frac{f}{f_c}\right)}{1 + j\left(\frac{f}{f_c}\right)} \quad ; \quad f_c = \frac{1}{2\pi RC} .$$

- (b) Determine the amplitude response of the phase shifter and, hence, explain how the phase shifter becomes an all-pass filter.
- (c) Determine the phase response and state the type of the phase shifter.
- (d) If $R_1 = 10 \text{ k}\Omega$, $R = 2 \text{ k}\Omega$ and $C = 0.01 \text{ }\mu\text{F}$, calculate the frequency of the input signal when the phase shift between input and output becomes 90° .

- 4) (a) The circuit diagram of a true integrator is shown in Figure 3. Show that the output is the integration of the input function by deriving an expression for the output voltage of the circuit.

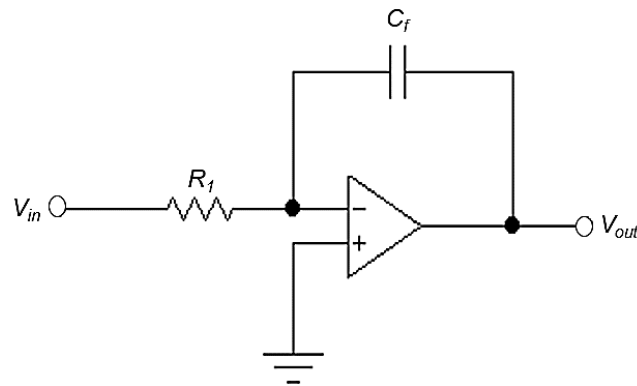


Figure 3

- (b) Sketch the output function of the true integrator if a constant DC voltage of 1 V is applied to the input at $t = 0$. Hence, find an expression for the gradient of the plot.
- (c) Derive the transfer function of the true integrator and discuss its main design issue in response to a very low-frequency input signals.
- (d) Modify the above true integrator to a practical integrator circuit to mitigate the above design issue and derive its transfer function.
- (e) Hence, show that the practical integrator has a low-pass frequency response.
- 5) (a) List five advantages of using active filters over passive filters.
- (b) Draw the first-order RC passive low-pass and high-pass filter circuits and derive their transfer functions. Hence, obtain expressions for the cut-off frequency of both filters.
- (c) Modify the above two circuits using an op-amp to have active gain control and obtain the transfer functions that you have derived in part (b) for modified circuits.
- (d) Sketch the frequency response curves for the modified filter circuits in part (c).
- (e) Design active (non-inverting) first-order low-pass and high-pass filter circuits having a cut-off frequency of 1 kHz. Use realistic component values.
Active filter gain (A_v) = 2.
- 6) (a) Describe the inverting and non-inverting modes of an analogue voltage comparator.
- (b) Using suitable diagrams, describe how an analogue voltage comparator becomes zero-level and voltage-threshold detectors.

- (c) Soil moisture is a measure of the water content of the soil. Since the soil solution is the medium for the supply of nutrients to the plants, soil moisture is a critical parameter for the plants' growth. In a given application, there are four plant growth fields, each with a water spraying system and soil moisture measuring sensor. Suppose the soil moisture level of a field is below the average soil moisture level of all four fields. In that case, the water spraying system of that field must be automatically turned ON and OFF when soil moisture goes up to the required level (average soil moisture level of all four fields).
- Design an op-amp-based circuit to measure the average of the sensor outputs of all four soil moisture sensors. Note that each sensor produces a positive voltage proportional to the soil moisture content of the respective field.
 - Design a circuit to control the soil moisture level of all four fields. Sensor outputs are the inputs of the circuit and outputs of the circuit switch ON and OFF the water spray system in each field.
- 7) (a) Define the terms "Line regulation" and Load regulation" in relation to the voltage regulation.
- (b) Determine the output voltage of the regulator shown in Figure 4.

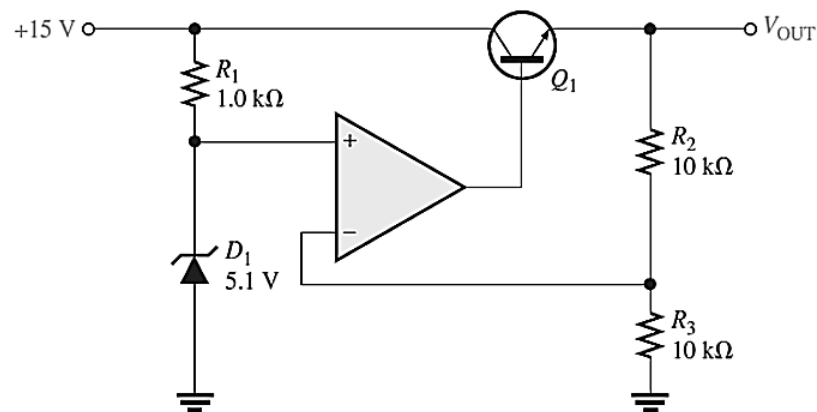


Figure 4

- Modify the regulator shown in Figure 4 to limit the maximum load current to be 700 mA.
- Draw the circuit diagram of a basic op-amp shunt regulator and explain its operation.
- Explain how a basic op-amp shunt regulator offers inherent short-circuit protection.
